

Polar Curves and Derivatives

ANSWER KEY



Converting between polar and rectangular coordinates

- 1 Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.
 $x = r \cos \theta = 4 \cos(\frac{\pi}{3}) = 4(\frac{1}{2}) = 2$. $y = r \sin \theta = 4 \sin(\frac{\pi}{3}) = 4(\frac{\sqrt{3}}{2}) = 2\sqrt{3}$. Point: $(2, 2\sqrt{3})$.
- 2 Convert the polar equation $r = 4 \cos \theta$ to rectangular form, and identify the curve.
Multiply by r : $r^2 = 4r \cos \theta \Rightarrow x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0 \Rightarrow (x - 2)^2 + y^2 = 4$. This is a circle of radius 2 centered at $(2, 0)$.

Identifying and sketching common polar curves

- 3 Identify the type of curve for $r = 2 + 2 \sin \theta$, and state its key feature.
This is a cardioid (since the coefficient of $\sin \theta$ equals the constant term) — a heart-shaped curve passing through the pole.
- 4 Identify the type of curve for $r = \cos(3\theta)$.
This is a rose curve with 3 petals (odd n in $r = \cos(n\theta)$ gives n petals).

The slope of a polar curve

- 5 State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r \cos \theta$ and $y = r \sin \theta$.
$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$
- 6 Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.
 $\frac{dr}{d\theta} = 2$. $x = 2\theta \cos \theta$, $y = 2\theta \sin \theta$. $\frac{dy}{d\theta} = 2 \sin \theta + 2\theta \cos \theta$; $\frac{dx}{d\theta} = 2 \cos \theta - 2\theta \sin \theta$. At $\theta = \frac{\pi}{2}$:
 $\frac{dy}{d\theta} = 2(1) + 2(\frac{\pi}{2})(0) = 2$; $\frac{dx}{d\theta} = 2(0) - 2(\frac{\pi}{2})(1) = -\pi$. $\frac{dy}{dx} = \frac{2}{-\pi} = -\frac{2}{\pi}$.

Finding horizontal and vertical tangents

- 7 For the circle $r = 4 \cos \theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r \sin \theta$ has a critical point).
 $y = 4 \cos \theta \sin \theta = 2 \sin(2\theta)$. $\frac{dy}{d\theta} = 4 \cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$.

Symmetry of polar curves

- 8 Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x -axis).
Replace θ with $-\theta$ in the equation. If the equation is unchanged (or yields the same curve), the curve is symmetric about the polar axis. For example, $r = 2 + 2\cos\theta$ satisfies $f(-\theta) = 2 + 2\cos(-\theta) = 2 + 2\cos\theta = f(\theta)$, confirming symmetry.
- 9 Test whether $r = 3\sin\theta$ is symmetric about the y -axis (replace θ with $\pi - \theta$).
 $f(\pi - \theta) = 3\sin(\pi - \theta) = 3\sin\theta = f(\theta)$ (using $\sin(\pi - \theta) = \sin\theta$). Symmetric about the y -axis.

More polar-to-rectangular conversions

- 10 Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \leq \theta < 2\pi$).
 $r = \sqrt{3^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$. $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ (in Quadrant I, matching both coordinates positive).
- 11 Convert the rectangular equation $y = x$ to polar form.
 $r\sin\theta = r\cos\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ (a constant- θ ray, i.e. a straight line through the pole).
- 12 Convert the polar equation $r\sin\theta = 4$ to rectangular form, and identify the curve.
 $r\sin\theta = y$, so $y = 4$: a horizontal line.

More slope-of-a-polar-curve practice

- 13 Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.
 $\frac{dr}{d\theta} = \cos\theta$. $\frac{dy}{dx} = \frac{\cos\theta\sin\theta + r\cos\theta}{\cos\theta\cos\theta - r\sin\theta}$. At $\theta = 0$: $r = 1$, $\frac{dr}{d\theta} = 1$. Numerator: $1(0) + 1(1) = 1$. Denominator: $1(1) - 1(0) = 1$. $\frac{dy}{dx} = \frac{1}{1} = 1$.
- 14 Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ , and interpret the result using the fact that a circle centered at the origin has radial symmetry.
 $\frac{dr}{d\theta} = 0$. $\frac{dy}{dx} = \frac{0 \cdot \sin\theta + 5\cos\theta}{0 \cdot \cos\theta - 5\sin\theta} = \frac{5\cos\theta}{-5\sin\theta} = -\cot\theta$. This matches the slope of the tangent line to a circle at angle θ — perpendicular to the radius at that point, consistent with basic circle geometry.

Tangent lines at the pole

- 15 Find the values of θ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.
 $1 - \cos\theta = 0 \Rightarrow \cos\theta = 1 \Rightarrow \theta = 0$. At the single angle where $r = 0$, the curve touches the pole; the tangent line there is along $\theta = 0$ (the polar axis), since that's the direction the curve approaches the origin from.

- 16** Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.
 $\cos(2\theta) = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — these are the four tangent-line directions at the pole for this 4-petaled rose.
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Classifying polar curves by equation type

- 17** Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):
- | | |
|---|--|
| a $r = 5$ Circle centered at the pole | b $r = 3 + 3\cos \theta$ Cardioid (coefficient equals constant) |
| c $r = 1 + 3\sin \theta$ Limaçon with an inner loop (constant < coefficient) | d $r = 4\sin(5\theta)$ Rose with 5 petals (odd $n = 5$) |
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Number of petals in rose curves

- 18** State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.
 If n is odd, the rose has exactly n petals (traced over $[0, \pi]$). If n is even, the rose has $2n$ petals (requires the full $[0, 2\pi]$ to trace, since each petal is retraced with opposite sign after π).
- 19** How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?
 $r = 2\sin(4\theta)$: $n = 4$ (even) $\Rightarrow 2(4) = 8$ petals. $r = 5\cos(7\theta)$: $n = 7$ (odd) $\Rightarrow 7$ petals.
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Graphing polar curves point-by-point

- 20** Complete a short table of (θ, r) values for $r = 2 + 2\cos \theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.
 $\theta = 0$: $r = 4$. $\theta = \frac{\pi}{2}$: $r = 2$. $\theta = \pi$: $r = 0$. $\theta = \frac{3\pi}{2}$: $r = 2$. The curve is widest ($r = 4$) along the polar axis, pinches to a point ($r = 0$) at $\theta = \pi$ (opposite direction), forming the characteristic heart shape.
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Concavity considerations for polar curves

- 21** Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).
 A vertical tangent means the slope $\frac{dy}{dx}$ is undefined (infinite) — since $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$, this happens exactly when the denominator $\frac{dx}{d\theta} = 0$ while the numerator stays nonzero, making the ratio blow up.